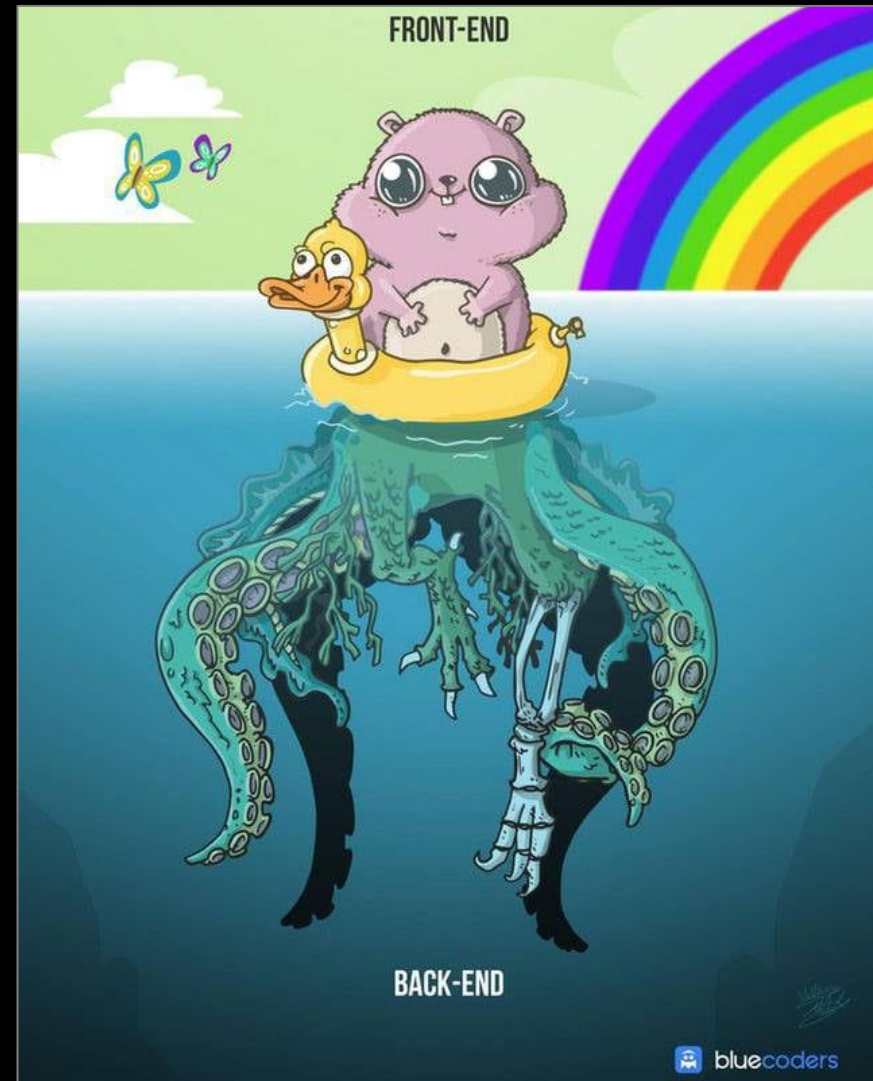


Web Technologies

**primary concepts
and vision**



Prof. Sabin Corneliu Buraga – profs.info.uaic.ro/sabin.buraga/
Assoc. Prof. Andrei Panu – profs.info.uaic.ro/andrei.panu/

“From a certain point onward,
there is no longer any turning back.
That is the point that must be reached.”

Franz Kafka

 what is the Web

 what does hypertext mean

 how do we identify Web resources

 what categories of Web applications can we develop

 what is the general architecture of a Web application



What is the Web?

world wide web

an Internet service

WWW $\neq$ Internet

world wide web

Common information space in which we communicate by
interconnecting and sharing elements of interest called
resources

invented by Sir **Tim Berners-Lee** in 1989

Sir Tim Berners-Lee, Inventor of the World Wide Web, to receive the ACM A.M. Turing Award

Sir Tim Berners-Lee – laureate of the Turing award (2017)
the highest distinction in Computer Science



<https://www.w3.org/> — 4 April 2017 — The [ACM](#), the Association for Computing Machinery, today named Sir Tim Berners-Lee, inventor of the Web and Director of the World Wide Web Consortium, as the recipient of the 2016 ACM A.M. Turing Award.

The [ACM Turing Award](#) is recognized as the highest distinction in Computer Science and is sometimes referred to as the "Nobel Prize of Computing." The Award is named for the British Computer Scientist [Alan Turing](#) who is known as the key founder of theoretical computer science and artificial intelligence as well as for the development of the Turing machine, considered a model of a general purpose computer.

Sir Tim is being given this award for inventing the World Wide Web, the first web browser, and the fundamental protocols and algorithms allowing the Web to scale. Considered one of the most influential computing innovations in history, the World Wide Web is the technical infrastructure of society and has already become the universal connectivity platform.

world wide web

Idea (Sir Tim Berners-Lee at CERN – 1989)

integrating distinct information systems
in a unitary manner,
without differences between data sources

world wide web

Idea (Sir Tim Berners-Lee at CERN – 1989)

integrating distinct **information systems**
in a unitary manner,
without differences between **data sources**

world wide web

Idea (Sir Tim Berners-Lee at CERN – 1989)

integrating distinct information systems
in a **unitary** manner,
without differences between data sources

world wide web

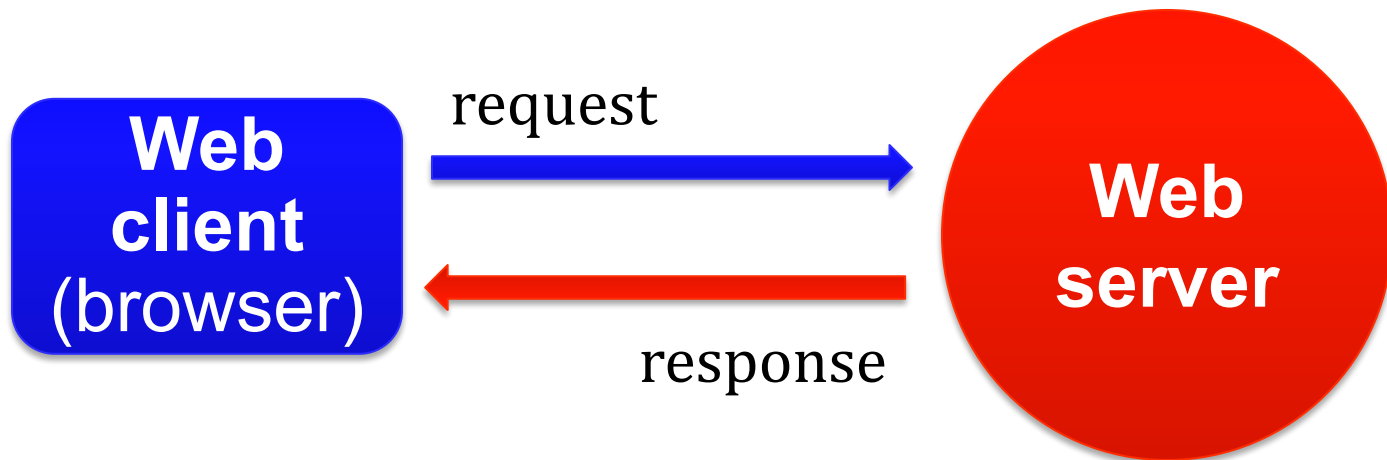
Idea (Sir Tim Berners-Lee at CERN – 1989)

integrating distinct information systems
in a unitary manner,
without differences between data sources

anything can link to anything

world wide web

based on **client/server** model

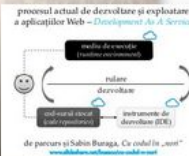


primul server Web

Etape în dezvoltarea unei aplicații Web

Cerințe – *requirements*
 Analiză și proiectare – *software design*
 Implementare – *build*
 Testare – *testing*
 Exploatare – *deployment*
 Mentenanță – *maintenance*
 Evoluție – *evolution*

Vreau să dezvolt un proiect Web...



◀ 14 of 118 ▶

1,383 views

Sabin Buraga: Dezvoltator Web?! (2019)

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Like

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...

Recommended



Web 2020 01/12: World Wide Web – aspecte arhitecturale
 Sabin Buraga



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 Sabin Buraga



Web 2020 04/12: Programare Web – Dezvoltarea aplicațiilor Web în PHP
 Sabin Buraga



Web 2020 05/12: Modelarea datelor. Familia XML. Extragerea datelor cu XPath.
 ...
 Sabin Buraga

...and on **hypertext** (hypermedia)

world wide web

Main intents:

device independence
software independence
scalability
ubiquity

world wide web

Based on open principles and standards
stipulated by the **Web Consortium**

Web for all & on everything

Web for rich interaction

Web of data & services

Web of trust

www.w3.org/mission/

Web architecture

Resources are identified by their **address**

URI – *Uniform Resource Identifier*

<https://profs.info.uaic.ro/andrei.panu/courses/web/syllabus.html>

Web architecture

Access to the Web resources' content
– resource representation –
is achieved by using a **protocol**

HTTP – HyperText Transfer Protocol

Web addresses
URI = URL + URN

Web protocols
HTTP, HTTPS, ...

data
encoding
Unicode

a b c ... z
ș 6 0 4 5 U □
: * ● 9

Internet
domain
names
DNS

.com .org .dev .ro
.uaic.ro
.info.uaic.ro

Internet
protocols
TCP/IP

TCP / UDP
IP, ICMP, ...
medium access

uniform resource
identifier

Web addresses
URI = URL + URN

secure HTTP
protocol

Web protocols
HTTP, HTTPS, ...

data
encoding
Unicode

a b c ... z
ș 6 0 4 6 U □
: * ● 9

Internet
domain
names
DNS

.com .org .dev .ro
.uaic.ro
.info.uaic.ro

Internet
protocols
TCP/IP

TCP / UDP
IP, ICMP,...
medium access

TCP = Transmission Control Protocol

IP = Internet Protocol

DNS = IP address ↔ domain name – e.g., **85.122.23.162** ↔ **profs.info.uaic.ro**

```

<body itemscope="" itemtype="http://schema.org/WebPage" id="top">

<div class="nav">
<nav>
<ul>
<li class="opt main"><a href="index.html" title="Tehnologii Web (spre pagina principal&#259;)" accesskey="W">#web</a></li>
<li class="opt film"><a href="web-film.html" title="Filmul desf&#259;ur&#351;rii materiei" accesskey="F">film</a></li>
<li class="opt projects"><a href="web-projects.html" title="Informa&#355;ii referitoare la proiecte"
accesskey="P">proiecte</a></li>
<li class="opt exam"><a href="web-exam.html" title="Am&#259;nunte privitoare la modul de examinare &#351;i notare"
accesskey="E">eval&nbsp;()</a></li>
<li class="opt biblio"><a href="web-biblio.html" title="Informa&#355;ii despre resursele bibliografice"
accesskey="B">biblio</a></li>
<li class="opt contact"><a href="web-contact.html" title="Posibilit&#259; &#355;i de contact (detalii viz&acirc;nd echipei
disciplinei)" accesskey="C">contact</a></li>
<li class="opt main"><a href="#top" title="La &icirc;nceputul acestei pagini Web" accesskey="S">&#x27F0;</a></li>
</ul>
</nav>
</div>

<article>
<header>
<h1 itemprop="name">Tehnologii Web</h1>
<p class="slogan" itemprop="description">resurse de interes</p>
</header>

<nav>
<p class="announce">
<a href="#books" title="Tip&#259;rituri">Tip&#259;rituri</a> &middot;
<a href="#web" title="Resurse Web">Resurse Web</a>
</p>
</nav>

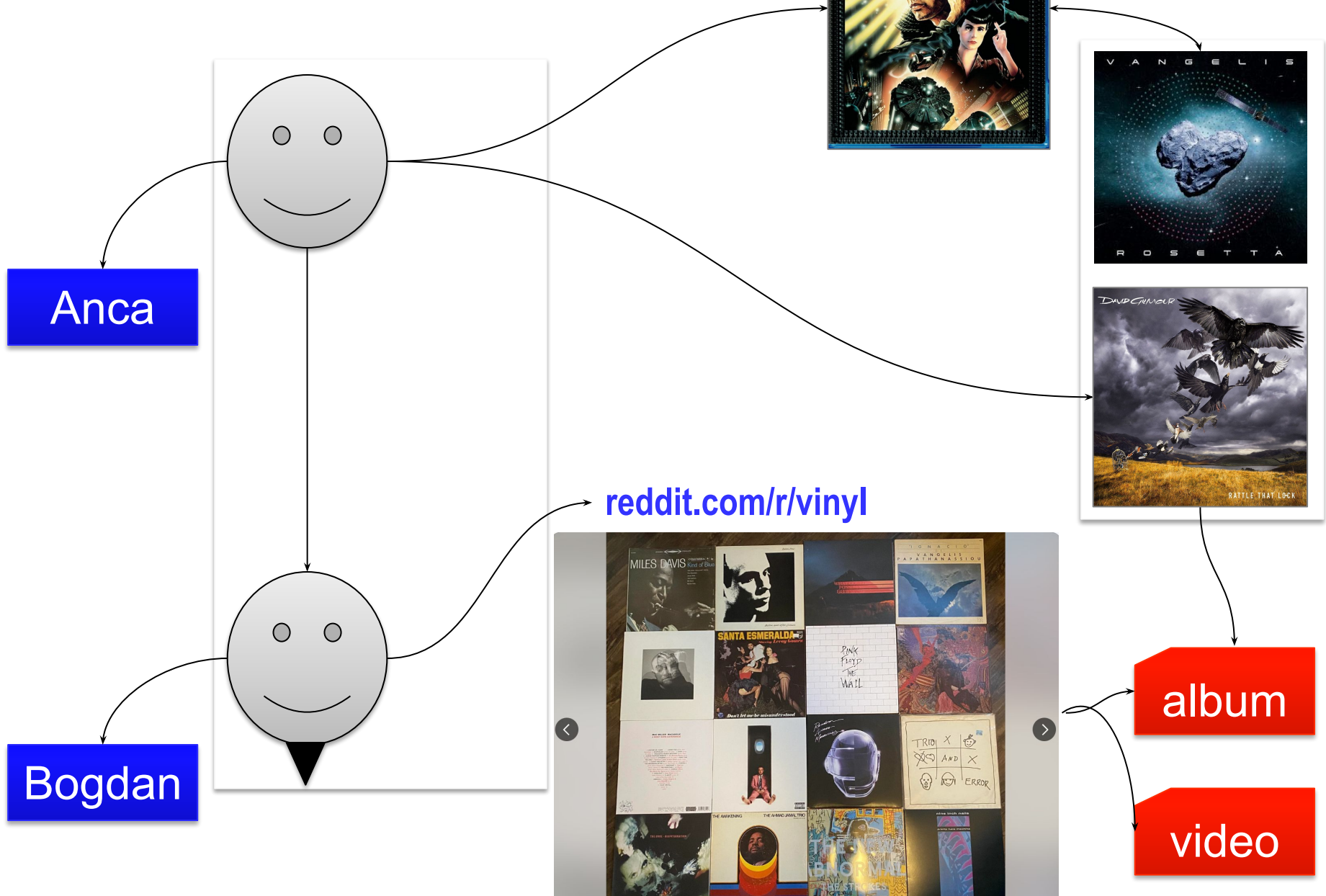
<section id="books">

```



resources – **documents** – include **<markups />**

markups themselves include
URIs ☐ **hypertext (hypermedia)**



Relations between a Web resource, its address (URI) and a structured representation of the resource

addressability via URI

<http://world.info/europe/romania/iasi/weather?today>

identifies

weather
forecast info
about Iași

Web resource

represents

representation

```
<section id="meteo">
  <div class="weather">
    <p lang="ro">Iași</p>
    <span>city</span>
    <p lang="en">Temp.
      <span id="today">
        is <strong>...</strong>
        &deg;C</span>
      </p>
    </div>
  </section>
```

HTML format
(typically, human users)

Relations between a Web resource, its address (URI) and a structured representation of the resource

addressability via URI

<http://world.info/europe/romania/iasi/weather?today>

identifies

weather
forecast info
about Iași

Web resource

represents

representation

```
<weather>
  <point lat="..." long="...">
    <name lang="ro">
      Iași
    </name>
    <type>city</type>
    ...
  </point>
  <temperature when="...">
    <value>...</value>
  </temperature>
</weather>
```

XML format
(software processable)

Relations between a Web resource, its address (URI) and a structured representation of the resource

addressability via URI

<http://world.info/europe/romania/iasi/weather?today>

identifies

weather
forecast info
about Iași

Web resource

represents

representation

```
{
  "point" : {
    "geo" : { "lat" : "...",
              "long" : "..."},
    "name" : "Iași",
    "type" : "city"
  },
  "temperature" : {
    "when" : "...",
    "value" : "..."
  }
}
```

JSON format
(software processable)

Relations between a Web resource, its address (URI) and a structured representation of the resource

addressability via URI

<http://world.info/europe/romania/iasi/weather?today>

identifies

weather
forecast info
about Iași

Web resource

represents

representation

```
<section id="meteo">
  <div class="weather">
```

representation

```
    <p lang="en">
      <span>
        <p lang="en">
          <span>
            ...
            "temperature" : {
              "when" : "...",
              "value" : "..."
            }
          ...
        </p>
      </div>
    </section>
```

representation – denoted by an open format (e.g., HTML, XML, JSON, RDF,...) – includes **data** itself + **meta-data**



What does hypertext mean?

hypertext: definition

“Written or graphical data,
interconnected in a complex manner,
which conventionally cannot be represented on paper.”

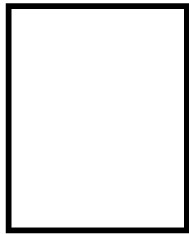
Ted Nelson, 1965

hypertext: definition

Non-linear text

hypertext: definition

Non-linear text



versus



hypertext: definition

A form of an electronic document

hypertext: definition

A form of an electronic document

open formats for content representation:

DocBook

HTML (HyperText Markup Language)

ODF (Open Document Format)

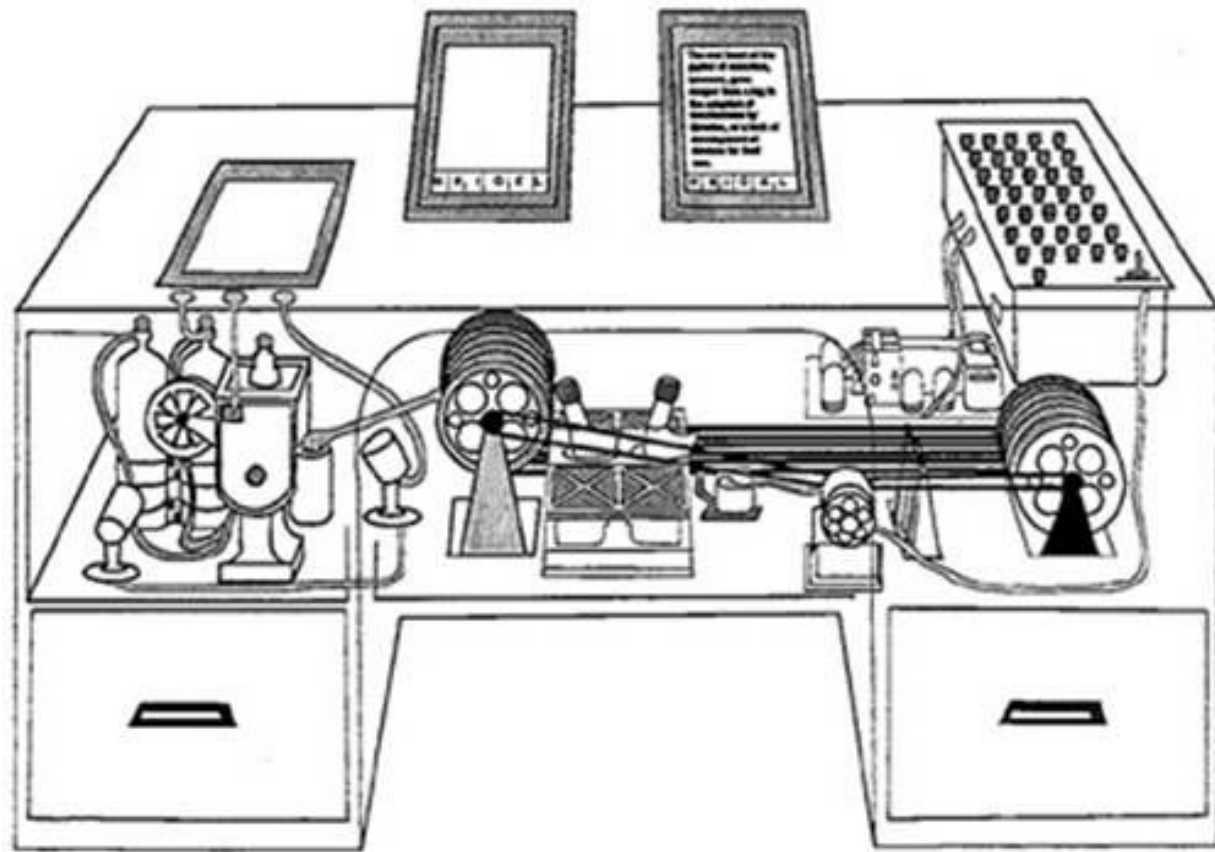
PDF (Portable Document Format)

...

hypertext: history

Vannevar Bush – *As We May Think*, 1945

MEMEX (MEMory EXtended)



“Consider **a future device for individual use**, which is a sort of mechanized private file and library. [...] It affords an immediate step, however, to associative indexing, the basic idea of which is a provision whereby any item may be caused at will to select immediately and automatically another. [...]

The process of tying two items together is the important thing.”

hypertext: history

Douglas Engelbart
Augment (1968)

mouse, graphical user-interface, text processing,
e-mail, scripting, windows on a screen, etc.

The Mother of All Demos (1968)

www.youtube.com/watch?v=yJDv-zdhzMY

hypertext: history

Ted Nelson

Xanadu – prototype, 1991

coined the “*hypertext*” term

*“Hypertexts: new forms of writing, appearing on computer screens,
that will branch or perform at the reader’s command.”*

*A hypertext is a non-sequential piece of writing;
only the computer display makes it practical.”*

archive.org/details/SelectedPapers1977/page/n7/mode/2up

hypertext: history

Hypermedia = **hypertext** + multimedia

Multimedia = Σ media

hypertext: history

Hypermedia = **hypertext** + multimedia

Multimedia = Σ media

communication media:
continuous (audio, video) and/or discrete (text)

hypertext: ingredients

Hypertext as a (di)graph

nodes = concepts

edges = relations

hypertext: ingredients

Nodes interconnected by links (edges)

source node = reference (anchor)

destination node = referent (anchor)

hypertext: links

referential
(non-hierarchical)

organizational
(hierarchical, structural)



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[Network](#)

[Perf](#)

[Personalization](#)

[Productivity](#)

[Testing](#)

[Webcomponents](#)

mdn

Learn > Core learning module

Filter

Learn web development

MDN curriculum

Getting started modules

Environment setup

Your first website

Web standards

Soft skills

Core modules

Structuring content with HTML

Basic HTML syntax

Web page metadata

Headings and paragraphs

Emphasis and importance

Lists

Test: HTML text basics

Advanced text features

HTML

CSS

JavaScript

Web APIs

All

Learn

Tools

About

Blog

Learn web development

FRONTEND DEVELOPER COURSE

Getting started modules

Core modules

MDN Curriculum

LEARN HTML

LEARN CSS

LEARN JAVASCRIPT

Structuring content with HTML module

CSS styling basics module

CSS layout module

Dynamic scripting with JavaScript module

Theme

English (US)

In this article

Prerequisites

Tutorials and challenges

Test your skills

Additional tutorials

See also

Prerequisites

Before starting this module, you don't need any previous HTML knowledge, but you should have at least basic familiarity with using computers and using the web passively (i.e., just looking at it and consuming content). You should have a basic work environment set up (as detailed in [Installing basic software](#)), and understand how to create and manage files (as detailed in [Dealing with files](#)). Both are parts of our [Getting started with the web](#) complete beginner's module.

Note:

If you are working on a computer, tablet, or another device where you can't create files, you can try running the code in an online editor such as [CodePen](#) or [JSFiddle](#).

Tutorials and challenges

[Basic HTML syntax](#)

static (indicated by the author of a document) *versus*
dynamic – generated by software

hypertext: documents

content

media types: text, image, audio, video,...

organization

nodes + structural links

presentation

textual, graphical, multimedia,
3D, mixed - (non-)interactive

(instead of) break



**FULL SNACK
DEVELOPER**



How are the Web resources
identified (addressed)?

URI

Each Web resource is denoted by
uniform resource identifiers

Uniform Resource Identifier

RFC 2396, 3986, 8820

datatracker.ietf.org/doc/html/rfc3986

URI: definitions

Resource

“thing” having an identity

URI: definitions

Resource

“thing” having an identity

note, CV, photo, presentation, song, program,
person, database, arbitrary concept, etc.

URI: definitions

Identifier

object that can denote a resource

URI: definitions

Identifier

object that can denote a resource

a character sequence (string)
conforming to a given syntax

URI: definitions

Uniformity

heterogeneous resources can be denoted by using the same syntactic conventions, being semantically interpreted in a uniform manner

$$\text{URI} = \text{URL} + \text{URN}$$

Uniform Resource Locator

identifies resources through the access mechanism:
network address, symbolic Internet domain

URI = URL + URN

Uniform Resource Locator

<http://dlinkrouter.local>

<https://profs.info.uaic.ro/andrei.panu/courses/>

<mailto:tux@penguin.info>

<ftp://ftp.funet.fi/pub/README.txt>

<data:image/png;base64,iVBORw0KGgoAA...YII=>

<tel:+40232201090>

<geo:47.16667,27.60000>

$$\text{URI} = \text{URL} + \text{URN}$$

Uniform Resource Name

identifies resources by name, in a persistent way,
even if the resource is an abstract one

URI = URL + URN

urn:mimetypes

urn:ISBN:973-681-988-4

urn:ietf:rfc:7700

urn:mozilla:install-manifest

URI = URL + URN

book uniquely
identified by
ISBN

urn:mimetypes ...

MIME
data types

urn:ISBN:973-681-988-4

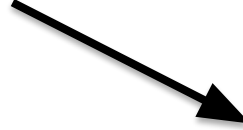
specification
(standard)

urn:ietf:rfc:7700 ...

software
component

urn:mozilla:install-manifest

schema://authority/path?query



<http://www.penguin.info/prog/search?id=Tux>

schema://authority/path?query



http://www.penguin.info/prog/search?id=Tux

schema represents a standardized addressing schema
(recognized by the client – *e.g.*, Web browser)

**about data file ftp geo http https im imap ipp ldap mailto
nfs sip sms stun tel turn tv urn ws xmpp** etc.

www.iana.org/assignments/uri-schemes/uri-schemes.xhtml

schema://authority/path?query



<http://www.penguin.info/prog/search?id=Tux>

authority can include user authentication information
(name:password – specified in plain text!)

+

data about the Internet domain/address,
optionally the port number

schema://authority/path?query



<http://www.penguin.info/prog/search?id=Tux>

path refers a (virtual) directory path towards a resource name – considered as a file name, optionally having an extension

schema://authority/path?query



<http://www.penguin.info/prog/search?id=Tux>

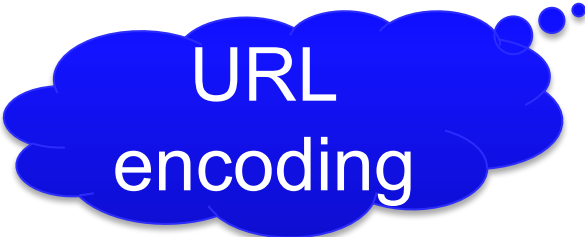
query specifies input data
usually, key=value pairs delimited by „&”

URI

Reserved characters

; / ? : @ & = + \$,

encoded in base 16, prefixed by %



URL
encoding

URI

Reserved characters

; / ? : @ & = + \$,

encoded in base 16, prefixed by %

example:

space will become %20



why?

<https://emojipedia.org/emoji/%F0%9F%96%A4/>

URI

Absolute URIs

<https://hypermedia.systems/hypermedia-a-reintroduction/>

it is mandatory to specify
schema and *authority* components

URI

Relative URIs

`../../web.css`

only the constructs regarding the *path*
– and, optionally, *query* – are indicated

URI

“Fragments” of a given content
can be referred by **URIref**
(references, fragment identifiers)

URI#URIref

URI

“Fragments” of a given content
can be referred by **URIref**
(references, fragment identifiers)

URI#URIref

web-biblio.html#web

<https://html.spec.whatwg.org/multipage/dom.html#documents>

<https://drive.google.com/#my-drive>

URI

Link directly to a text fragment
in the absence of a URlref identifier

URI#:~:text=[prefix-,]textStart[,textEnd][,-sufix]

details:

developer.mozilla.org/docs/Web/URI/Reference/Fragment/Text_fragments

URI

Link directly to a text fragment
in the absence of a URlref identifier

example: using

<https://edu.info.uaic.ro/web-technologies/web-film.html#:~:tex>

t=HTML in the Web browser,

this will highlight the first occurrence of the text "HTML"

- Resurse: **The History of the Web, The birth of the Web** (CERN)
- Laborator: **Limbajul de marcare HTML**
 - Resurse: **Every HTML Element, HTML Tutorials** (MDN), **HTML Exercises, HTML First Best Practices** 💰 , **DevTools Tips** 💰

URI

URIs must be considered **opaque**

the content type must not be “guessed”
by inspecting the URI indicating a resource

URI

URIs must be considered **opaque**

the content type must not be “guessed”
by inspecting the URI indicating a resource

a resource’s type is not given by the extension – *e.g.*, **.html** –,
but by the MIME type (Media Type) indicated by the server



see the next
lecture

URI

URIs must be considered **opaque**

the content type must not be “guessed”
by inspecting the URI indicating a resource

the resource’s state/content can change in time,
but not its URI

“Cool URIs don’t change” – www.w3.org/Provider/Style/URI.html

IRI

Internationalized Resource Identifier

permits the use of Unicode characters

also see IDN (Internationalized Domain Name)

examples:

<https://thefreedictionary.com/rosé>

<https://dexonline.ro/definitie/păianjen> . .

<https://blog.twnic.tw/tag/勒索軟體/>

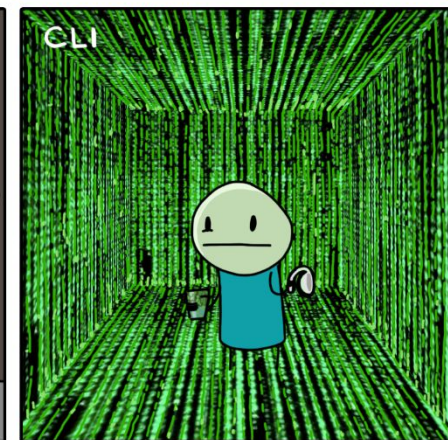
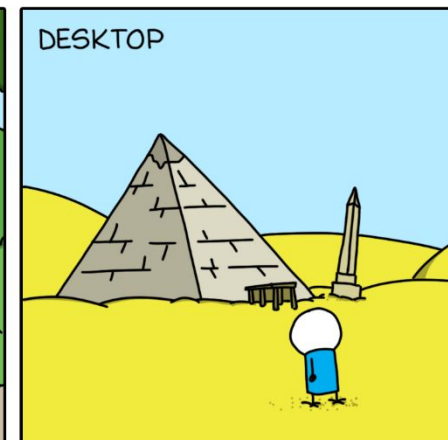
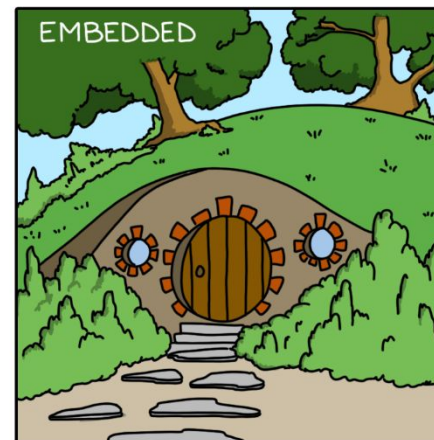
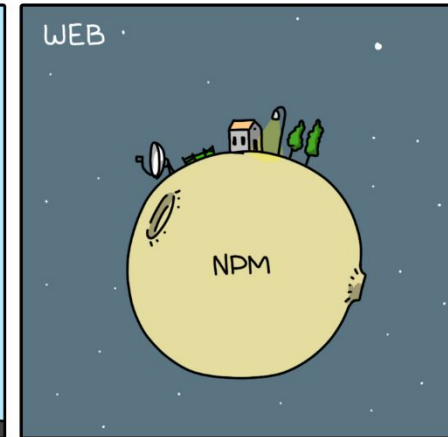
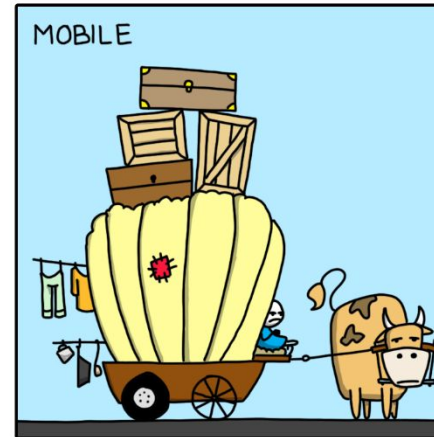
p%C4%83ianjen



What kind of Web
applications
can we develop?

APP TYPES AS HOMES

MONKEYUSER.COM



Web site

a system on which a Web server is running,
hosting a set of related pages (resources)

of an organization, company, or person(s)

Web application

a collection of interconnected Web pages having the
content dynamically generated,
in order to offer a specific functionality to its users

Web application

the interaction between application and users
is facilitated by a Web interface

Web application

the interaction between application and users
is facilitated by a Web interface

usually, Web site $\equiv$ Web application

Web application

examples:

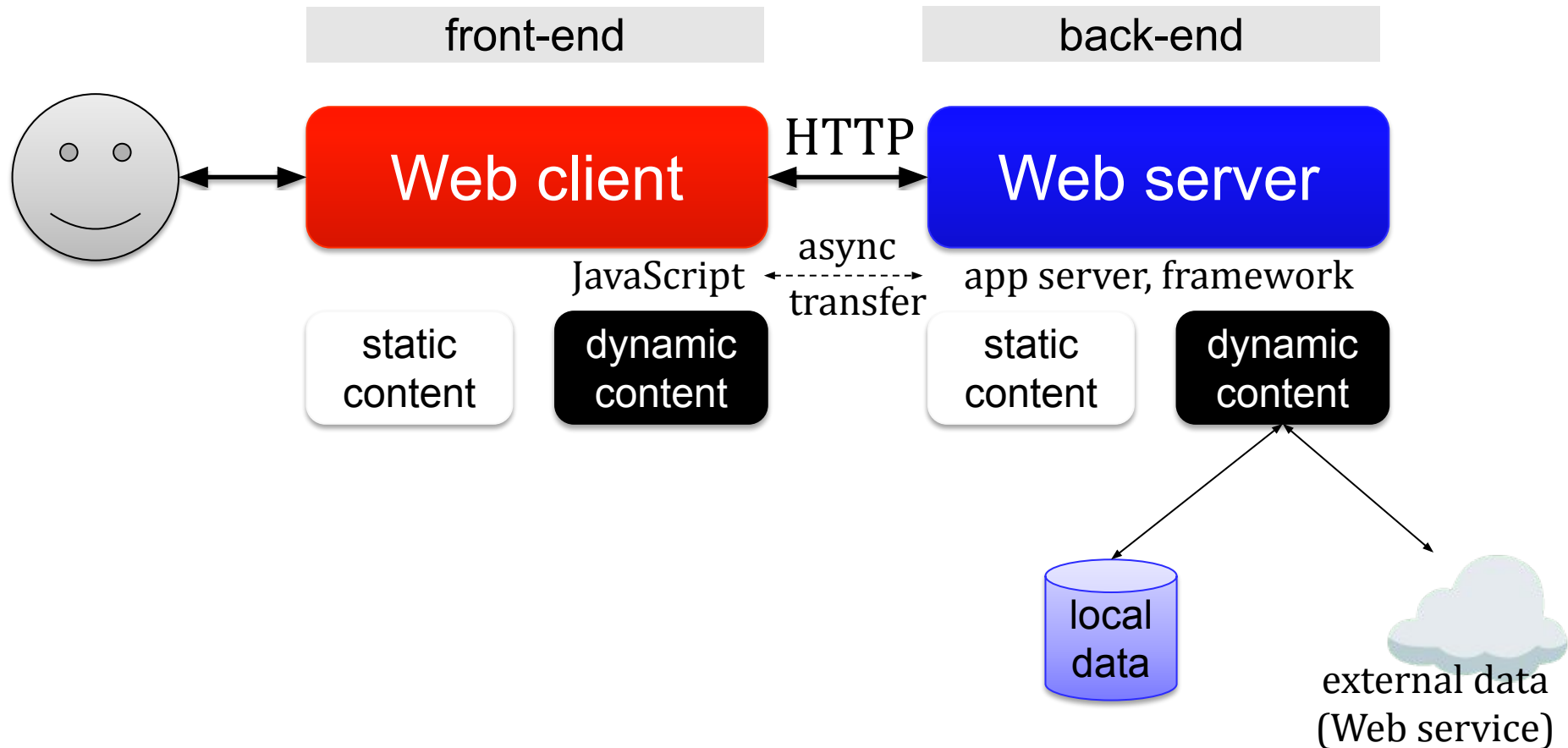
Coursera, Devdocs.io, DuckDuckGo, Europeana, Figma, GitHub, Grid.Space, Heroku, info.uaic.ro, InfoQ, JSBin, Kongregate, Medium, OpenStreetMap, Pocket, Reddit, Replit, Spliit, Synth.Town, Unsplash, WikiArt, WordPress



...and many, many, many others



What is the architecture of a
Web application?



via a **Web interface**, the user interacts with the client (**front-end**) and initiates actions – e.g., (a)synchronous HTTP requests – which will be executed by various **components** implemented at the server level (**back-end**), in order to obtain **data**

Web app. = Interface + Program + Content (Data)

Web app. = Interface + Program + Content (Data)

myth 1: the most important thing is the interface

myth 2: the most important thing is the program

myth 3: the most important thing is the data

open standards:

HTML, CSS, Ajax, SVG, WebGL, WebXR, WAI...

Web app. = Interface + Program + Content (Data)

myth 1: the most important thing is the interface

myth 2: the most important thing is the program

myth 3: the most important thing is the data

server: C#, Java, JavaScript, PHP, Python, Ruby, Scala,...

client: JavaScript, WebAssembly (WASM)

Web app. = Interface + Program + Content (Data)

myth 1: the most important thing is the interface

myth 2: the most important thing is the program

myth 3: the most important thing is the data

relational (SQL), graph (NoSQL), JSON, XML, RDF, etc.

A. Gavra, *Frameworks for Understanding Databases* (2025):

www.bitsxpages.com/p/frameworks-for-understanding-databases

Web app. = Interface + Program + Content (Data)

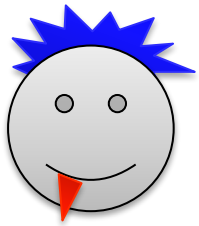
myth 1: the most important thing is the interface

myth 2: the most important thing is the program

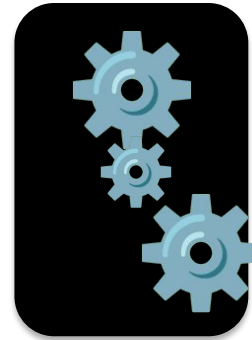
myth 3: the most important thing is the data

fact: **all of them are important!**

Client
user
interface

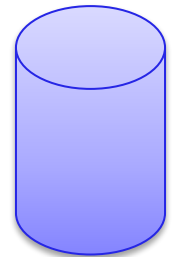


Internet
(Web)



Server
Web site/application

Data
stored
persistently



Web Application = Interface + Program + Content (Data)

Web applications

complex software systems,
in a permanent evolution

Reality

multiple ways of Web interaction with the user



mobile



laptop



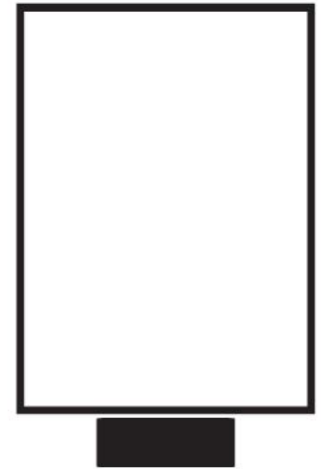
PC



tablet



(*smart*) TV



urban screen

plus, newcomers:



game console



smart watch



smart clothing



smart appliances



smart home



smart

Reality

an increased number of **users**,
having greater expectations from actual software

from (hyper-)text content 
to social Web applications   + natural interaction
  

Reality

the support for **developing applications**
(languages, APIs, libraries, tools,...)
provided by the hardware/software platform
on server-side and/or client-side

Reality

lack of adjustment
regarding the economic (business) requirements



development vs.



marketing vs.



management

Reality

with respect to large Web projects

launch delays
under budgeting
lack of functionality
poor application quality

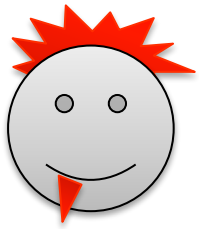
goals
psychology
behavior

interaction
controls
language

features
technology
algorithms

indexing
structure
meta-data

tools
methodologies
incentives



users

interface

software

content

creators

adapted from Crumlish & Malone, 2009

“conslusions”

“One of the things the Web teaches us is that everything is connected (hyperlinks) and we all should work together (standards). Too often school teaches us that everything is separate (many different 'subjects') and that we should all work alone.”

Aaron Swartz

summary



terminology, client/server model, hypertext,
URI, site vs. application, Web apps categories

500

Internal Server Error

YOUR PERFECTLY VALID REQUEST



www.agiliq.com/notebook/http-status-memes

next episode:

Web programming

from HTTP to
CGI scripts



302

Found

http.cat